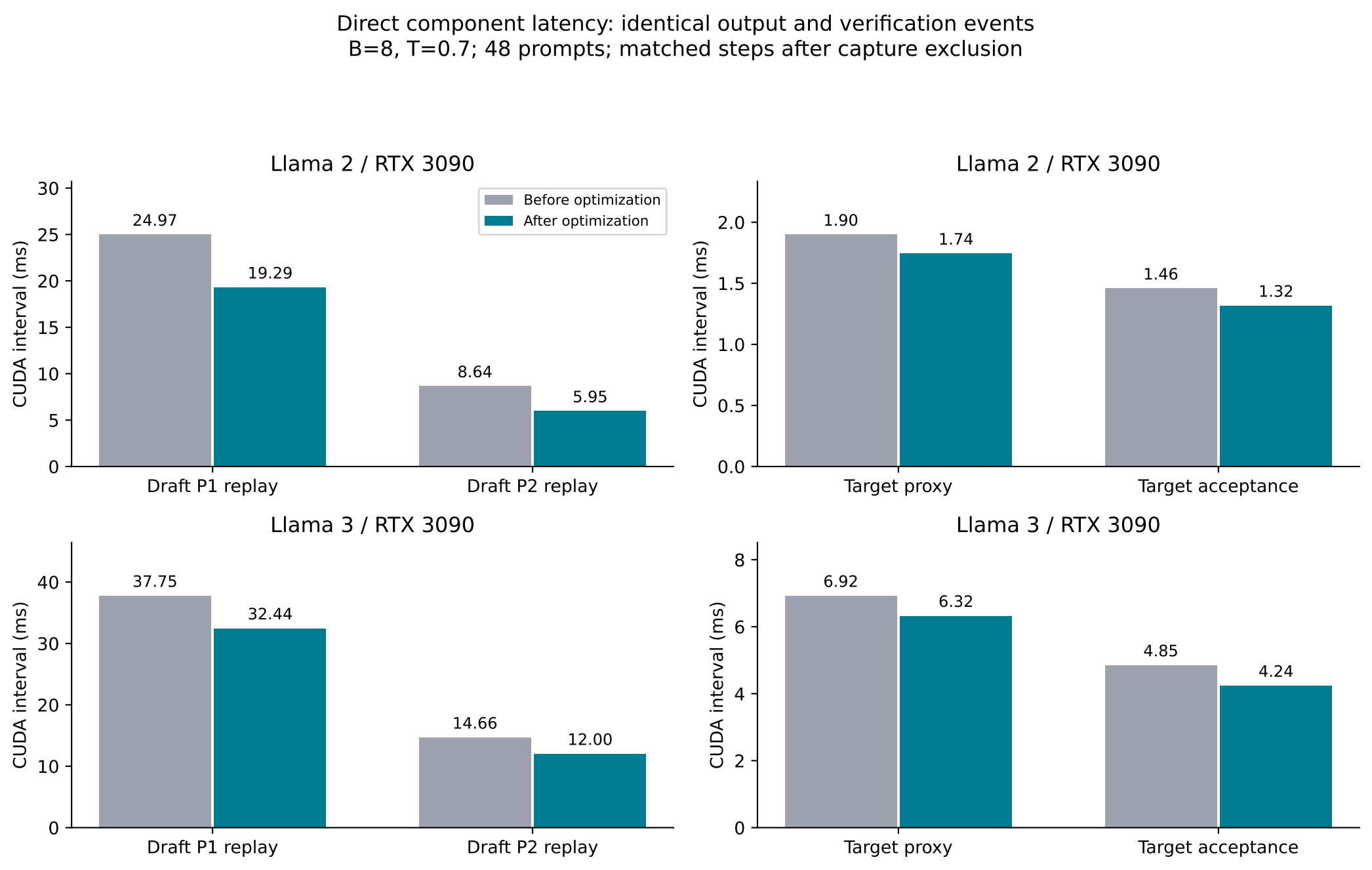
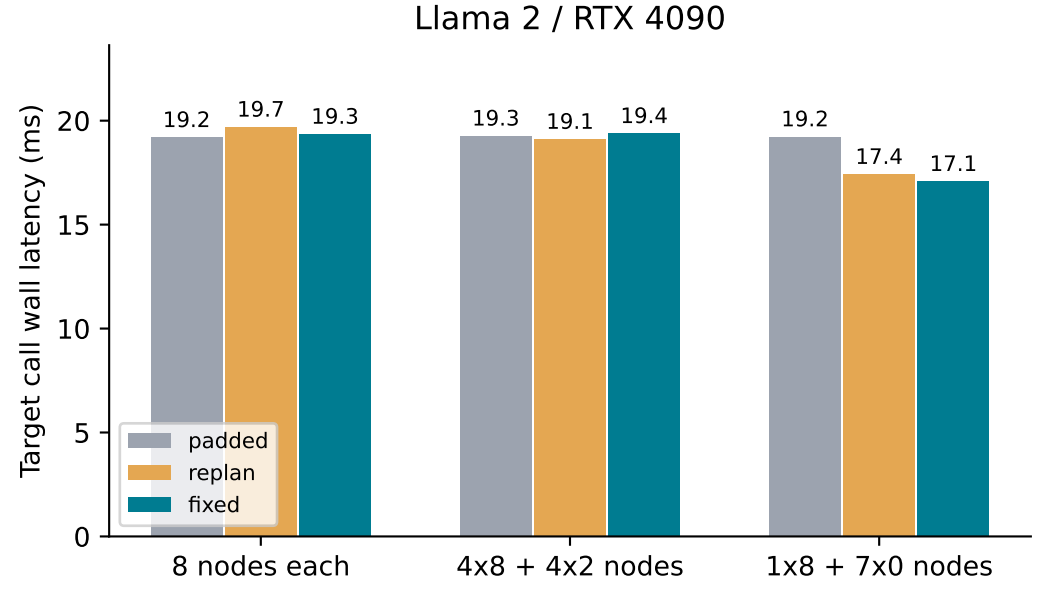
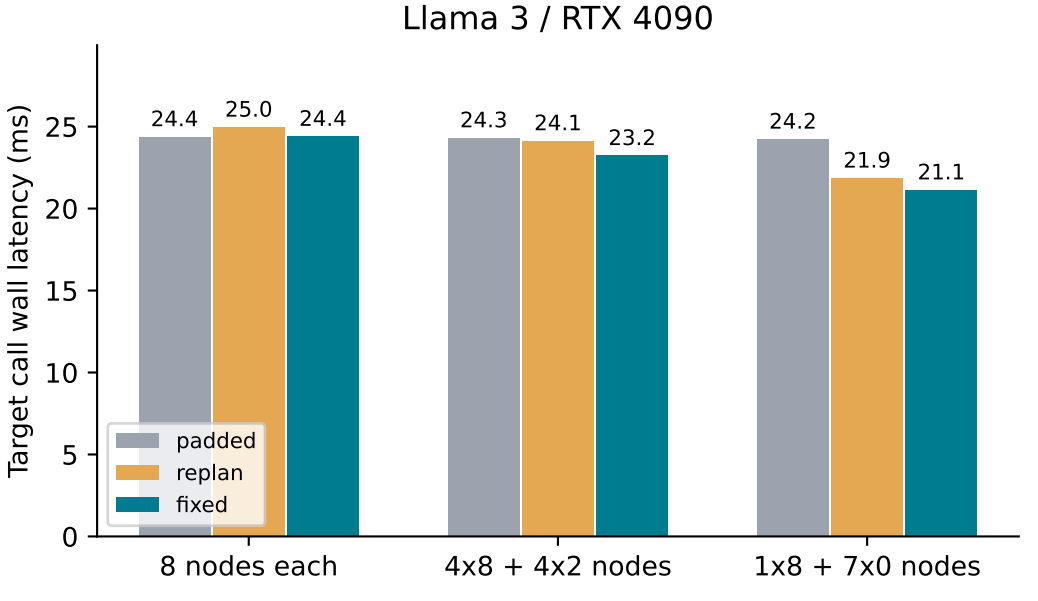
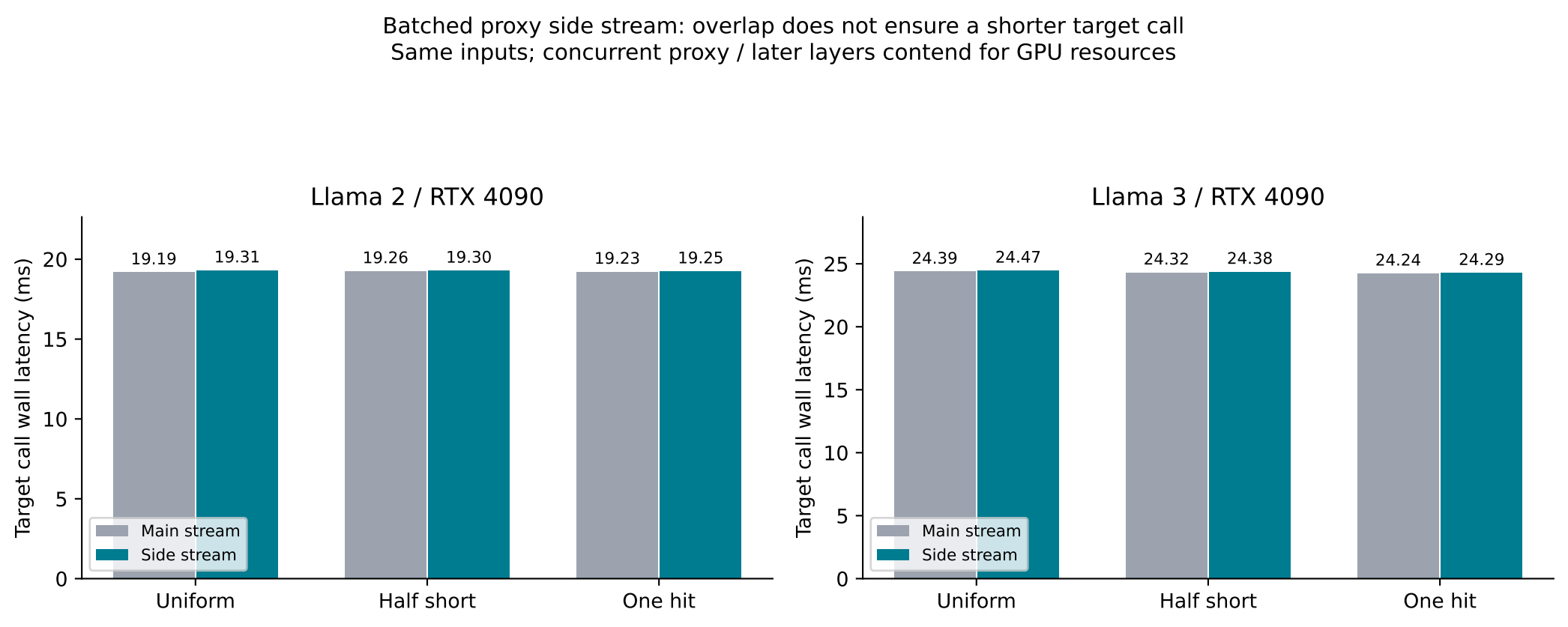
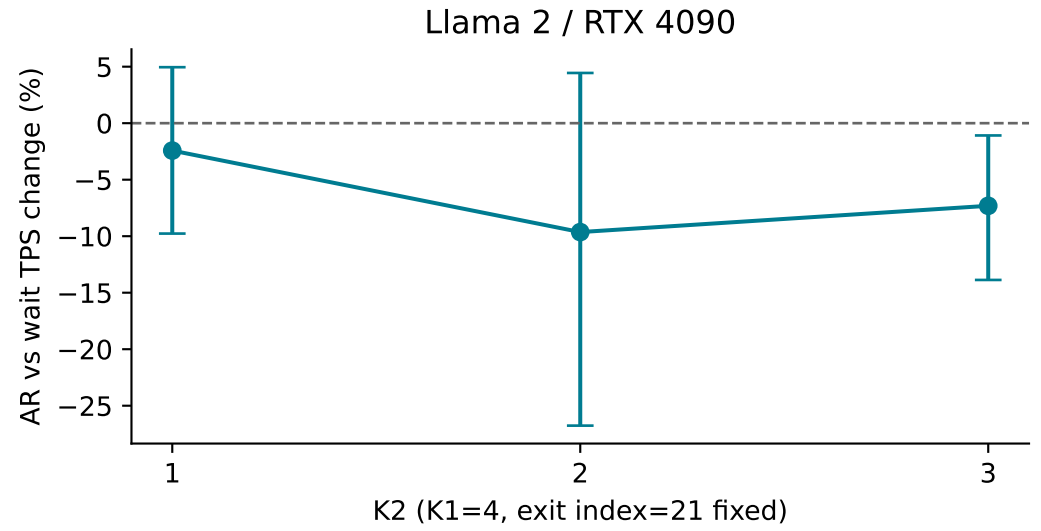
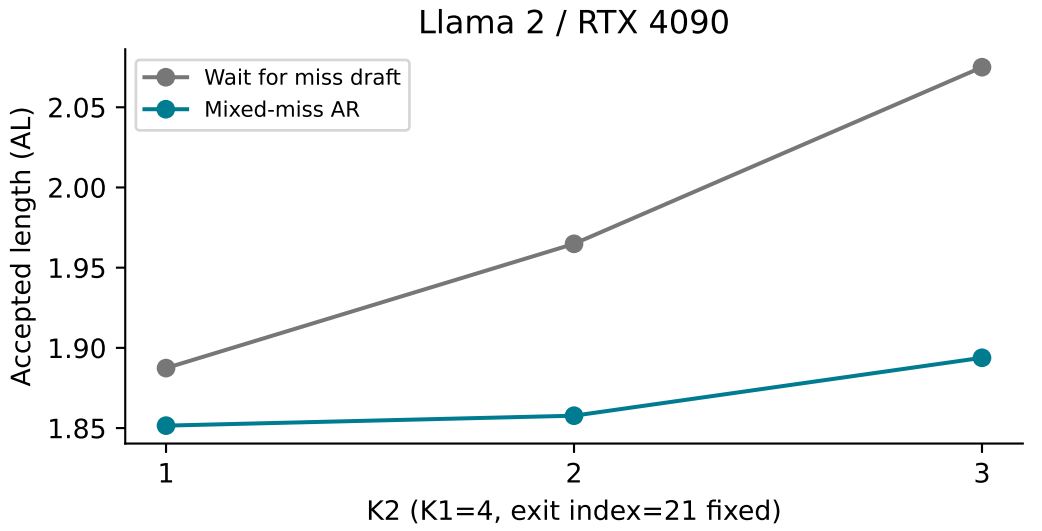
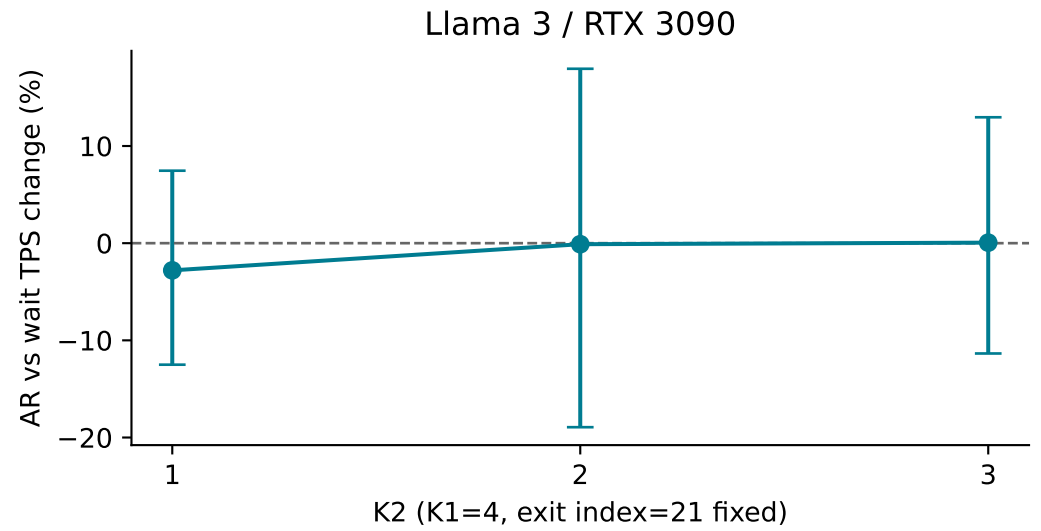
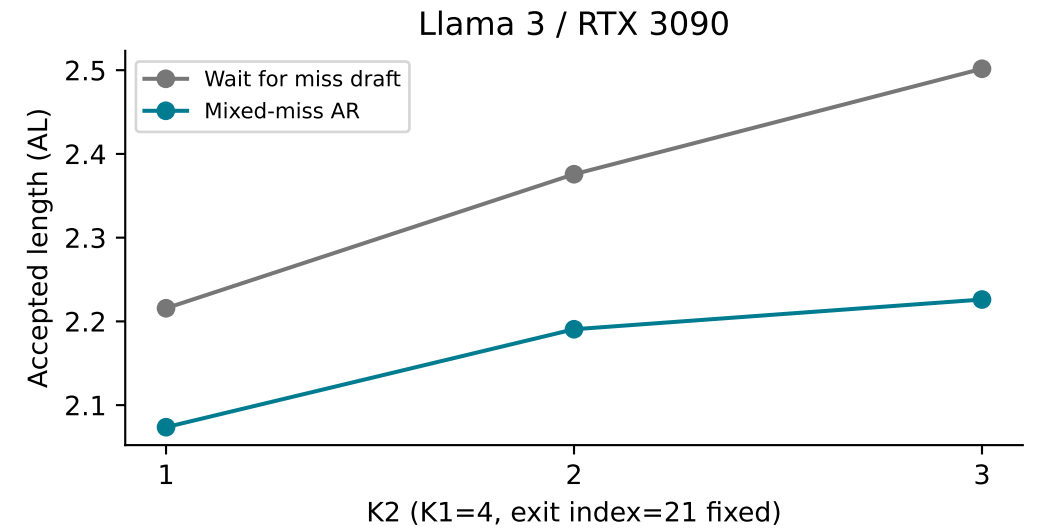

Direct component latency: identical output and verification events
B=8, T=0.7; 48 prompts; matched steps after capture exclusion
Llama 2 / RTX 3090
Before optimization
After optimization
CUDA interval (ms)
24.97
19.29
8.64
5.95
Draft P1 replay
Draft P2 replay
Llama 2 / RTX 3090
1.90
1.74
1.46
1.32
Target proxy
Target acceptance
Llama 3 / RTX 3090
37.75
32.44
14.66
12.00
Draft P1 replay
Draft P2 replay
Llama 3 / RTX 3090
6.92
6.32
4.85
4.24
Target proxy
Target acceptance

Packed tree: same synthetic shape, real model + prefilled KV
Steady replay, 20 repeats; includes proxy, excludes draft. Numerical outputs may differ.






Batched proxy side stream: overlap does not ensure a shorter target call
Same inputs; concurrent proxy / later layers contend for GPU resources
Llama 2 / RTX 4090
Llama 3 / RTX 4090
Target call wall latency (ms)
Main stream
Side stream
19.19
19.31
19.26
19.30
19.23
19.25
24.39
24.47
24.32
24.38
24.24
24.29
Uniform
Half short
One hit

Mixed-miss policy sensitivity: TPS and AL together
480 prompts per cell; paired 95% bootstrap CI over 60 prompt groups; not optimal tuning